



Open Notion Hub

FinStream

Real-Time Financial Intelligence Platform

Project workspace
Notion Engineering Hub



Click or scan: docs, roles, timeline, execution plan

"From raw transaction to real-time decision in milliseconds."

10,000+

Transactions / sec

<100ms

Fraud Latency Target

6

Engineers

3

Real-life Problems

Meet the Team

FinStream · Team of 6 Engineers

E1

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E2

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Executive Summary

FinStream is an end-to-end, production-grade data engineering platform that processes high-velocity financial events through real-time and batch pipelines, stores them in a medallion lakehouse, and surfaces actionable fraud detection, risk scoring, and market intelligence.

Real-Time Fraud Detection

Every transaction scored for fraud within <100ms using live ML models embedded in the Flink pipeline.

Dynamic Risk Scoring

Risk profiles updated per transaction — not per day — using rolling aggregates stored in Redis.

Market Intelligence

Live market signals from Polygon.io, news sentiment, and FX data processed into decisioning signals.

Full Audit & Observability

Immutable Bronze layer, Prometheus metrics, Grafana dashboards, and Great Expectations quality gates.

FinStream turns raw financial events into decisions while they are happening.

Real-time streams and batch Lakehouse features connect to fraud, risk, and market intelligence use cases.



pitch: FinStream turns raw financial events into real-time fraud decisions, dynamic risk profiles, and market intelligence signals using a governed streaming lakehouse architecture.

Delayed financial decision-making is the business problem.

WHAT WE SOLVE

Each source is mapped to an explicit operational decision, not just stored for tooling practice.

Business Problem	Why It Matters	FinStream Response
Fraud is detected too late	Money may already be transferred before fraud is found	Score every transaction in real time using Flink plus ML/rules
Risk changes continuously	Daily batch risk scores become outdated	Update profiles from live behavior plus historical Gold features
Market/news events move exposure quickly	Price drops, negative sentiment, or volatility create risk	Convert ticks and sentiment into market intelligence signals
Raw data is scattered	Analysts and ML teams cannot trust inconsistent sources	Store events in governed Bronze/Silver/Gold lakehouse layers
Users need live and historical insight	Dashboards must show current alerts and trends	Serve real-time and curated data through ClickHouse/Superset

Every ingested dataset earns its place by producing a decision.

DATA LOGIC

This answers why transactions, market data, and news belong in the same financial data platform.

Data We Ingest	What We Extract	Decision / Output
Bank transactions	Amount patterns, velocity, merchant behavior, country/device anomalies	Fraud score, risk level, approve/review/block decision
Reference data	Customer profile, card history, merchant risk, country risk	Better fraud scoring and dynamic risk scoring
Historical transactions	30/60/90-day behavior, fraud rate, average amount, abnormal patterns	Gold features for ML training and risk profiles
Market ticks / FX / price data	Price spikes, volume anomalies, volatility	Market signals and exposure alerts
News / sentiment feeds	Sentiment plus topic/entity signals	Market intelligence and optional risk context
Raw Bronze event history	Immutable audit trail	Replay, debugging, compliance, model retraining

The architecture converts streaming events into live decisions and trusted Gold features.

DECISION FLOW

Real-time scoring and batch enrichment share the same governed lakehouse foundation.



The technologies form one decision system across fraud, risk, and market intelligence.

USE-CASE MATRIX

Each layer contributes a specific business capability and avoids tool sprawl.

Capability	Fraud Detection	Dynamic Risk	Market Intelligence	Chosen Tools
Live event capture	Every transaction arrives as an event	Behavior changes arrive continuously	Ticks/news arrive continuously	Kafka + Schema Registry
Real-time scoring	Score transaction before settlement	Update risk after behavior changes	Detect volatility/sentiment signals	Flink + Redis
Audit + replay	Replay suspicious cases	Rebuild profiles from history	Replay market/news windows	MinIO + Iceberg
Historical features	Fraud rates, velocity, averages	30/60/90-day profiles	Volatility and sentiment history	Spark + dbt
Dashboards	Fraud alerts and outcomes	Risk profile trends	Market/exposure signals	ClickHouse + Superset
Production trust	SLAs, quality gates, run history	Reliable feature publishing	Healthy jobs and metrics	Airflow + Docker + GX + Prom/Grafana

Project Description & Core Idea

What is FinStream?

FinStream is a production-grade, end-to-end data engineering platform designed to ingest, process, store, and serve high-velocity financial and transactional data in real time.

It combines a streaming speed layer (Apache Flink), a batch enrichment layer (Apache Spark), and a unified medallion lakehouse (MinIO + Apache Iceberg) to deliver a single source of truth for fraud detection, risk scoring, and market intelligence.

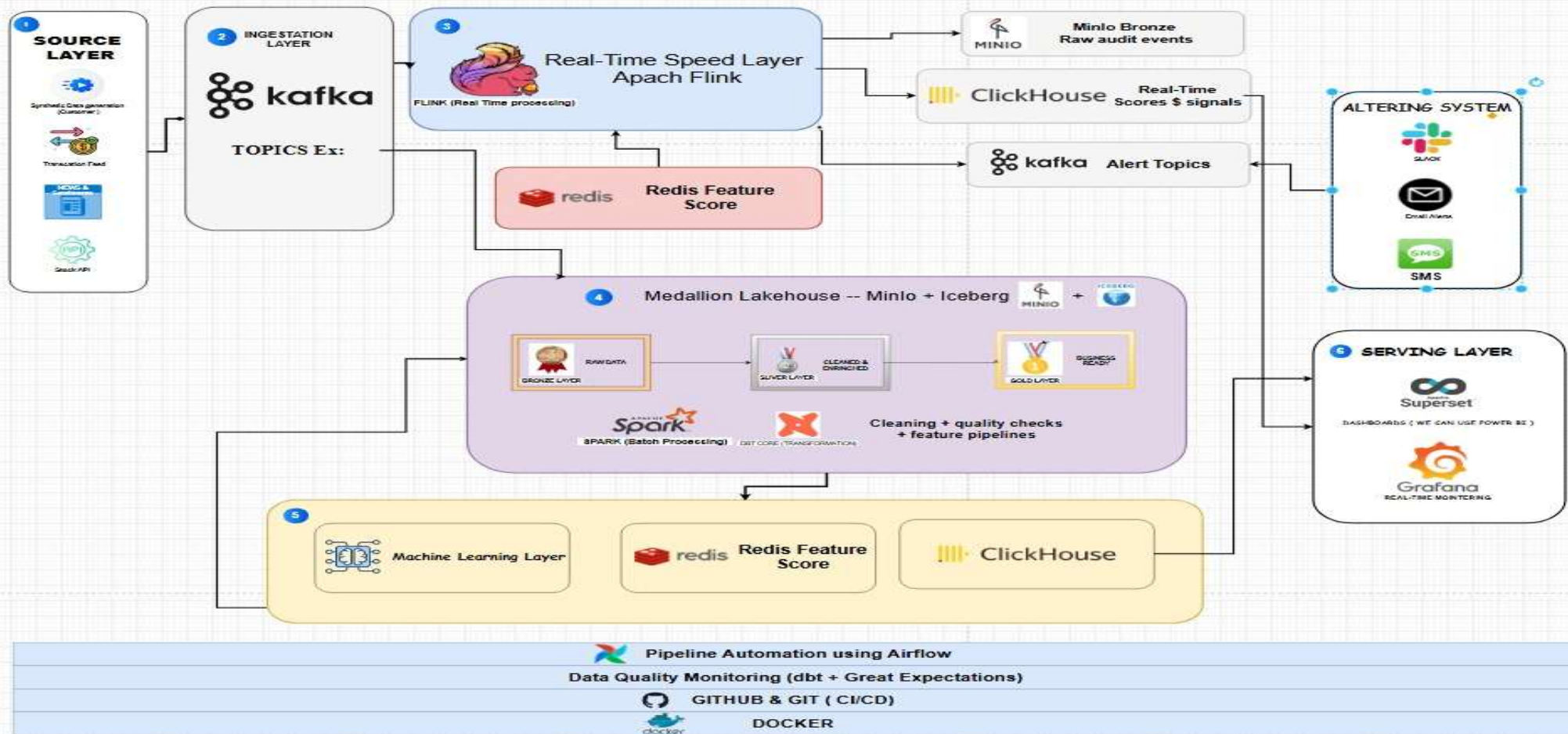
The platform processes synthetic bank transactions, live market ticks, and social sentiment feeds simultaneously, making it a realistic simulation of an enterprise-grade financial intelligence system.

The Core Idea

- 1 Traditional financial platforms process data in hourly or daily batches — fraud is detected only after money is already moved. Bad Risk Scoring and far from market analysis
- 2 FinStream eliminates this lag using a Lambda Architecture: a real-time speed layer (Flink) and a batch layer (Spark) running in parallel.
- 3 Every transaction is scored within milliseconds using ML models trained on historical Gold data and served live inside Flink.
- 4 Three Flink output paths run simultaneously: raw storage to MinIO Bronze, live scores to ClickHouse, and high-risk alerts to Kafka.
- 5 The medallion lakehouse (Bronze → Silver → Gold) ensures all consumers share one governed, high-quality source of truth.

Data Flow Diagram "Sketch Using Draw.io"

FOR CLAIRTY PURPOSE



Tech Stack – Tools



Benefits & Target Outcomes

Business Outcomes

Fraud Detection

Transactions scored and flagged before they complete settlement.

Dynamic Risk Scoring

Risk profiles updated on every event — not on a daily batch schedule.

Market Intelligence Signals

Real-time decisioning signals from prices, FX, and sentiment data.

Immutable Audit Trail

Every raw event preserved permanently in the Bronze layer for compliance.

Automated Actions

Card blocking and Slack/email alerts triggered without human intervention.

Technical / Portfolio Outcomes

Full-Stack Data Engineering

Streaming, batch, lakehouse, MLOps, and DevOps — in one coherent system.

Production-Grade Patterns

Exactly-once semantics, schema evolution, data quality contracts, model governance.

Enterprise-Transferable Stack

Every tool is industry-standard and in active use at Fortune 500 companies.

CI/CD Driven Platform

GitHub Actions, Docker, and Airflow deliver automated testing and deployment.

End-to-End Observability

Prometheus metrics and Grafana dashboards cover every layer of the platform.

Datasets & Sources

Dataset	Source	Format	Volume
Synthetic Transactions	Transaction events	600,000 events/min (10,000/sec)	300–600 MB/min
IEEE-CIS Historical Fraud Data	Batch training only	Not streamed	Used offline
News & Sentiment (Live)	Articles / sentiment events	8–15 events/min	1–5 MB/min
UCI Customer Data	Reference updates	<1 event/min	Negligible
Historical News (GDELT)	Batch ingestion	Not streamed	Used for model training
Live Market News	News headlines/events	5–20 events/min	1–10 MB/min
Historical Stock Market Data	Batch ingestion	Not streamed	Used for back testing

All datasets are either open-source, freely accessible via API, or synthetically generated for portfolio and educational use.

Pipeline Sequence — Steps 1 to 5

1 of 2

#	Step Name	Layer	Tools	Description
1	Data Generation	Source Layer	Python, Faker, Polygon.io, NewsAPI, Reddit API 	Python scripts simulate 10K tx/sec. Market ticks pulled from Polygon.io. News sentiment fetched from NewsAPI and Reddit. Acts as the live event source.
2	Event Ingestion	Ingestion Layer	Apache Kafka, Confluent Schema Registry, Kafka Producer 	Events published to 5 Kafka topics. Schema Registry validates all messages against Avro schemas. Bad messages routed to a Dead Letter Queue topic.
3	Stream Processing	Real-Time Layer	Apache Flink, Redis, PyFlink  	Flink validates, enriches, and scores events in milliseconds. Redis provides sub-ms feature lookups. Outputs fan out to 3 parallel paths simultaneously.
3a	Path 1 — Raw Store	Flink Output	Flink Sink, MinIO, Parquet / Avro  	All raw events written immutably to MinIO Bronze layer the moment Flink processes them. Creates permanent audit record for compliance and replay.
3b	Path 2 — Analytics	Flink Output	Flink Sink, ClickHouse 	Real-time fraud scores and market signals written directly to ClickHouse. Enables sub-second dashboard queries without waiting for batch jobs.

Pipeline Sequence — Steps 6 to 9

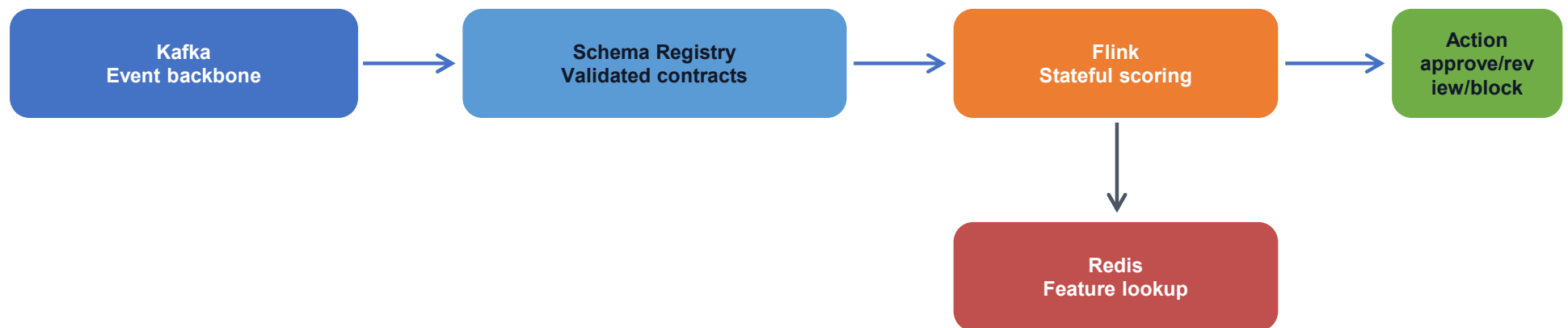
2 of 2

#	Step Name	Layer	Tools	Description
3c	Path 3 — Alerts	Flink Output	Flink Sink, Kafka fraud_alerts	High-risk events emitted to the fraud_alerts Kafka topic. Downstream consumers read this topic to trigger automated alerting and card-blocking actions.
4	Lakehouse Storage	Storage Layer	MinIO, Apache Iceberg, dbt Core  	Bronze → Silver → Gold medallion on MinIO with Iceberg as open table format. dbt Core manages SQL transformation logic between tiers.
5	Batch Processing	Batch Layer	Apache Spark, dbt Core, Airflow  	Airflow-scheduled Spark jobs build customer profiles, risk features, and Gold marts. Great Expectations validates quality at each tier boundary.
6	ML Training	ML Layer	MLflow, scikit-learn, XGBoost, Feature Store 	Models trained on Silver/Gold features, tracked in MLflow. Registered models deployed live into the Flink scoring pipeline via the Model Registry.
7	Serving & Analytics	Serving Layer	ClickHouse, Apache Superset, Grafana  	ClickHouse serves real-time and historical OLAP queries. Superset for dashboards. Grafana for system health. APIs expose scores to applications.
8	Alerting System	Alerting Layer	Python consumer, SMTP, Slack API, Card Management API	Python consumer reads fraud_alerts topic and dispatches email, Slack, and SMS notifications. Automated card-blocking via mock API.
9	Infrastructure	DevOps Layer	Docker, Airflow, GitHub Actions, Prometheus, Grafana 	All services containerized in Docker. GitHub Actions CI/CD. Airflow orchestrates batch jobs. Prometheus + Grafana cover full observability.

Kafka, Flink, and Redis fit the fraud decision path.

SPEED LAYER

This is the part of FinStream that must act while the transaction is still fresh.



Tool	Use Case Fit	Advantages
Kafka	Transactions, market ticks, news	Durable event stream; decouples producers from processors and sinks.
Schema Registry	Financial event contracts	Prevents silent schema drift across topics.
Flink	Fraud + risk scoring	Stateful event-time windows, joins, and low-latency enrichment.
Redis	Online features	Fast customer/card/merchant risk lookup during scoring.

Sources: Apache Kafka introduction; Apache Flink stateful stream processing docs; Redis docs.

Distribution of Work — Team of 6

E1

Data Ingestion Engineer

Python data producers, Kafka topic design & partitioning, Schema Registry setup, DLQ monitoring

Python · Kafka · Avro · Schema Registry

E2

Stream Processing Engineer

All 3 Flink output paths, Redis feature store integration, real-time fraud/risk scoring, stateful ops & exactly-once semantics

Apache Flink · Redis · PyFlink · Java

E3

Lakehouse & Transformation Eng.

MinIO setup, Iceberg table design, dbt models (Bronze→Silver→Gold), Great Expectations quality checks

MinIO · Iceberg · dbt Core · SQL

E4

Batch Processing & Analytics Eng.

Spark batch jobs, ClickHouse schema & ingestion, Superset dashboard development, ad-hoc SQL analytics

Spark · ClickHouse · SQL · Superset

E5

ML & Feature Engineering

Feature pipelines from Silver/Gold, model training & evaluation, MLflow tracking, model registry, live Flink inference integration

Python · MLflow · scikit-learn · XGBoost

E6

Platform Engineering & DevOps

Docker Compose setup, GitHub Actions CI/CD, Airflow DAGs, Prometheus & Grafana monitoring, alerting system

Docker · Airflow · GitHub Actions · Grafana

Project Category

Big Data

Qualifies on all three Vs: Volume (10K tx/sec), Velocity (sub-second processing), and Variety (transactions, market ticks, news, FX).

Lambda Architecture

Speed layer (Flink real-time) + batch layer (Spark) run in parallel and reconcile in ClickHouse — the textbook Lambda pattern.

Lakehouse Architecture

MinIO + Apache Iceberg + Bronze/Silver/Gold medallion pattern combines data lake flexibility with data warehouse governance.

MLOps Platform

MLflow tracking, feature store, model registry, and live Flink inference make this a full MLOps system, not just a data pipeline.

DataOps / Platform Eng.

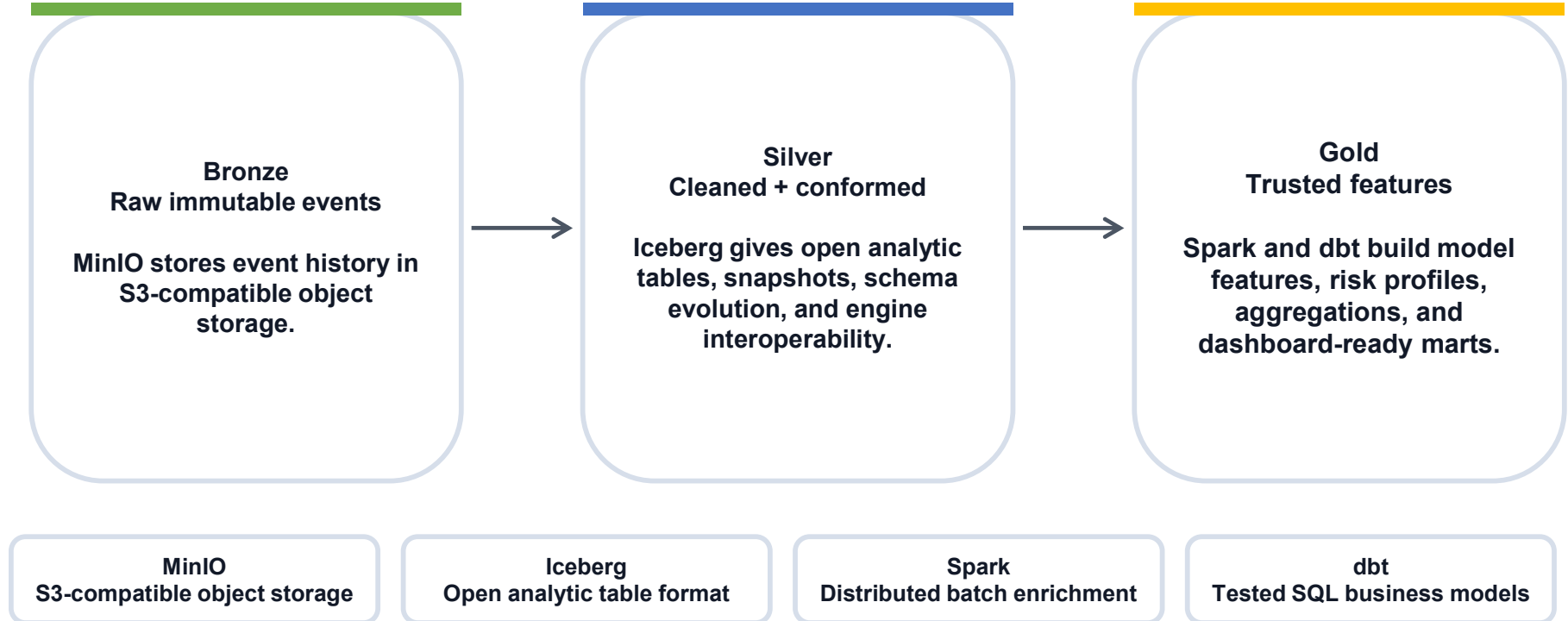
CI/CD with GitHub Actions, Airflow orchestration, Great Expectations quality gates, and full Prometheus + Grafana observability.

Portfolio positioning: A production-grade Lakehouse + Lambda architecture with an embedded MLOps layer, designed for real-time financial intelligence at enterprise scale.

MinIO, Iceberg, Spark, and dbt make the raw data trustworthy.

LAKEHOUSE LAYER

The goal is not only storage; it is replay, audit, feature building, and repeatable business logic.



Sources: MinIO docs, Apache Iceberg docs, Apache Spark docs, dbt documentation.

ClickHouse and Superset serve business users; Prometheus and Grafana serve engineers.

SERVING LAYER

FinStream needs two visibility surfaces: financial intelligence dashboards and platform health dashboards.

Business analytics

ClickHouse receives curated/live analytics tables for fast SQL over alerts, fraud outcomes, risk profiles, and market signals.

Superset turns those tables into analyst-facing dashboards.

ClickHouse
Real-time analytics

Superset
Business dashboards

Operational observability

Prometheus captures service and pipeline metrics: throughput, lag, failures, latency, and resource health.

Grafana visualizes whether the platform is meeting the <100ms target.

Prometheus
Pipeline metrics

Grafana
Ops dashboards

Airflow, Docker, and Great Expectations turn the project into a repeatable platform.

PRODUCTION CONTROL

This is what moves the proposal from a demo pipeline toward an enterprise-style data engineering system.

Airflow

Schedule and monitor batch workflows, backfills, dbt runs, quality checks, and training jobs.

Docker

Package services consistently so Kafka, Flink, Spark, MinIO, Redis, ClickHouse, and Superset run as one reproducible environment.

Thank You

FinStream — Real-Time Financial Intelligence Platform

- Real-time fraud detection at <100ms
- Medallion lakehouse with Iceberg + MinIO
- Lambda architecture: Flink + Spark
- Full MLOps with MLflow model registry
- CI/CD, Docker, Airflow, Prometheus
- Team of 6, clearly distributed ownership

Questions & Discussion



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